In the diagram 2.16, the indifference Curves IC_1, IC_2 and IC_3 represent the Indifference Map, Upper IC representing higher level of satisfaction compared to lower IC.

Marginal Rate of Substitution

The shape of an indifference curve provides useful information about preferences. Indifference curve replaces the concept of marginal utility with the concept of the marginal rate of substitution.

According to Leftwich "*The marginal rate of substitution of X for Y (MRS_{xy}) is defined as the maximum amount of Y the consumer is willing to give up for getting an additional unit of X and still remaining on the same indifference curve*".

2.13 Diminishing Marginal Rate of Substitution

It explains the concepts of diminishing marginal rate of substitution.

Since y decrease as X increases, the change in Y is negative i.e., $-\Delta y$, so the equation is

$$MRS_{xy} = -\frac{\Delta Y}{\Delta X} and$$

However, as with price elasticity of demand the convention is to ignore the minus sign in

$$MRS_{xy} = \frac{\Delta Y}{\Delta X}$$

2.14 Properties of the Indifference Curves

Indifference curves are subjective and unique to each person. Nevertheless they have in common the following properties:

1. **Indifference curve must have negative slope**

   An indifference curve has a negative slope, which denotes that if the quantity of commodity (y) decreases, the quantity of the other (x) must increase, if the

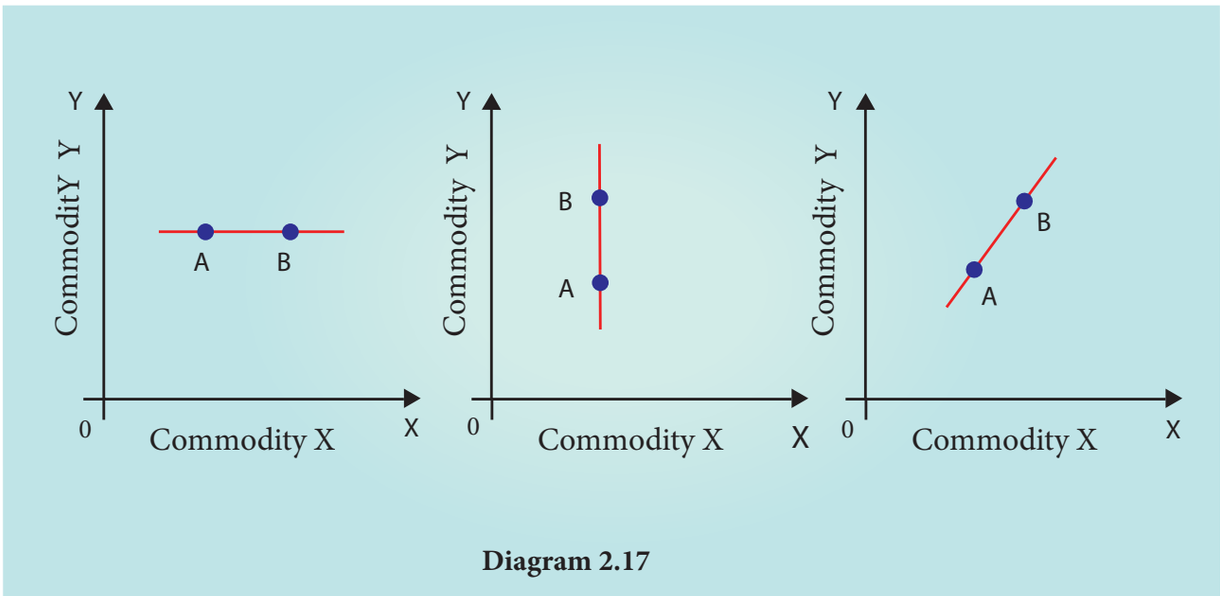


Diagram 2.17

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